

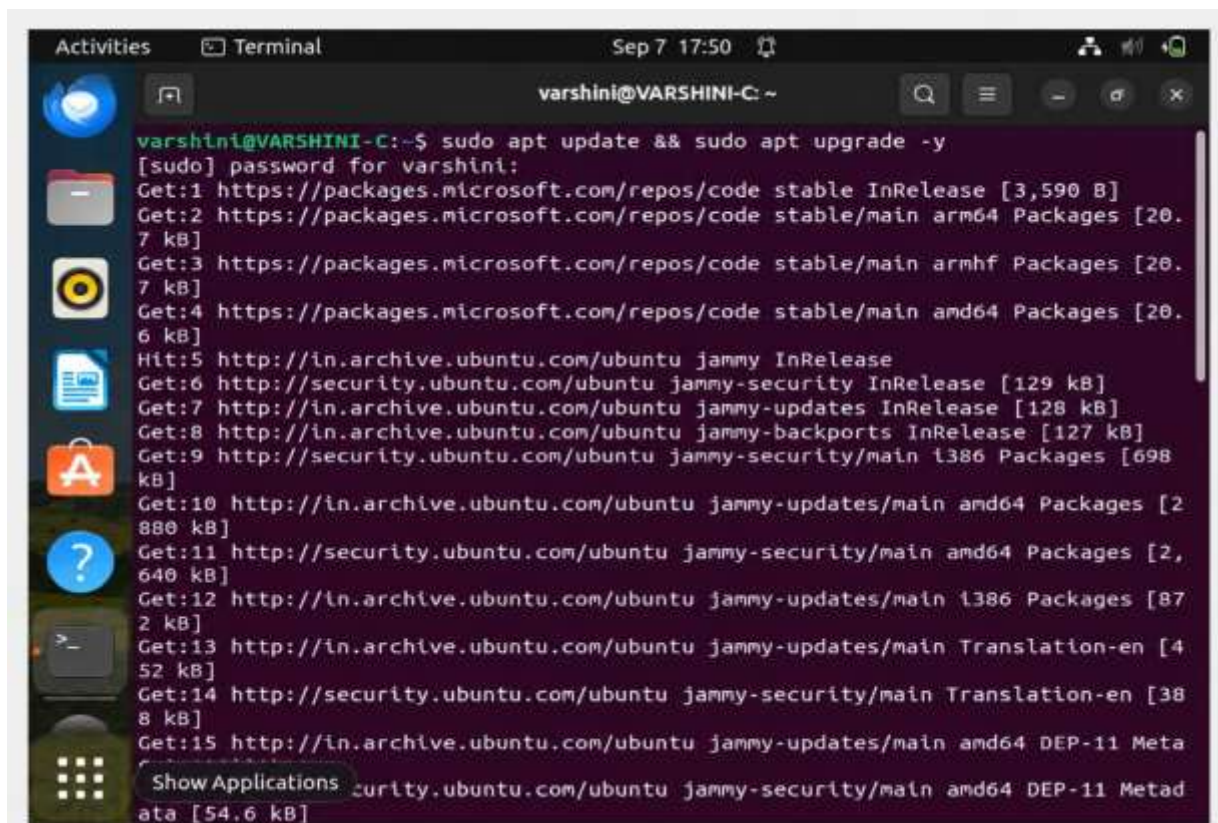
8) Installation of OpenStack and creation of Virtual machines

Aim:

To install of openstack and creation of virtual machine in linux system.

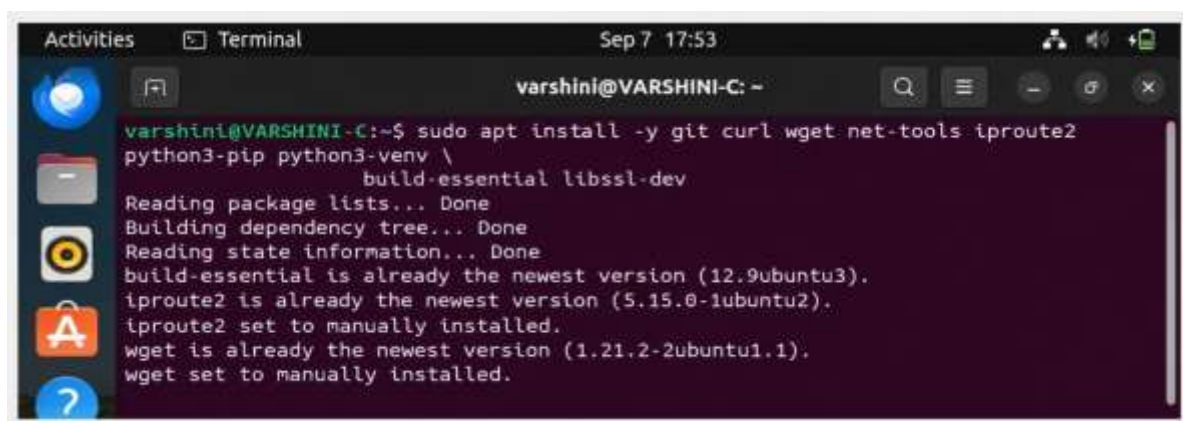
Steps:

1) Update all the file in the ubuntu by sudo apt update



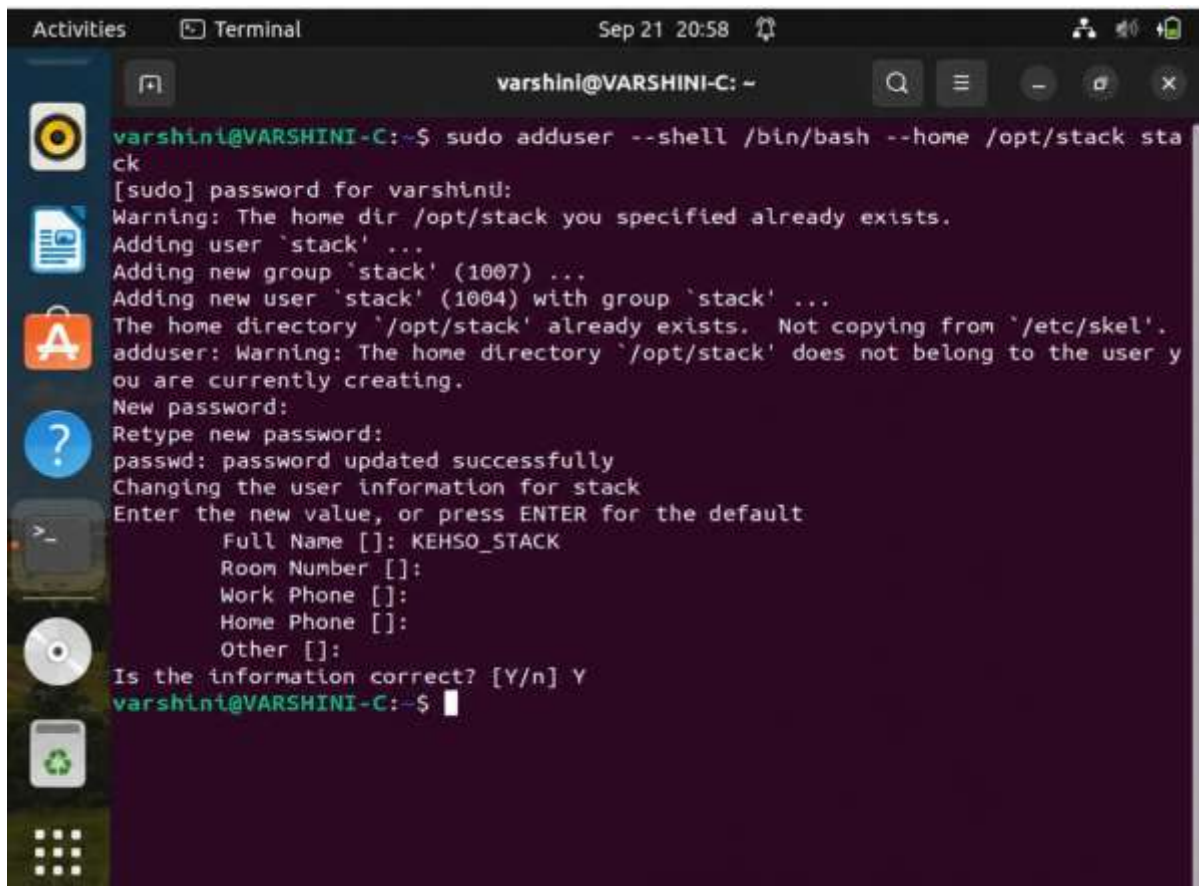
```
varshini@VARSHINI-C:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for varshini:
Get:1 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]
Get:2 https://packages.microsoft.com/repos/code stable/main arm64 Packages [20.7 kB]
Get:3 https://packages.microsoft.com/repos/code stable/main armhf Packages [20.7 kB]
Get:4 https://packages.microsoft.com/repos/code stable/main amd64 Packages [20.6 kB]
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:6 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [698 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2880 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2,640 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [872 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [452 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [388 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Meta
Show Applications
ata [54.6 kB]
```

2) Install the packages to build the openstack



```
varshini@VARSHINI-C:~$ sudo apt install -y git curl wget net-tools iproute2
python3-pip python3-venv \
    build-essential libssl-dev
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.9ubuntu3).
iproute2 is already the newest version (5.15.0-1ubuntu2).
iproute2 set to manually installed.
wget is already the newest version (1.21.2-2ubuntu1.1).
wget set to manually installed.
```

3) Add the user stack to the ubuntu especially for the openstack and give the sudo privilege for the user stack and add to the group

A terminal window titled 'varshini@VARSHINI-C: ~' showing the execution of the 'sudo adduser' command to create a new user named 'stack'. The user is added with a shell of '/bin/bash' and a home directory of '/opt/stack'. The terminal output shows the user being added, the group 'stack' being created, and the user's information being set. The user is then prompted to enter a password and confirm it. Finally, the user is prompted to enter their full name, room number, work phone, home phone, and other information. The user confirms the information is correct by pressing 'Y'.

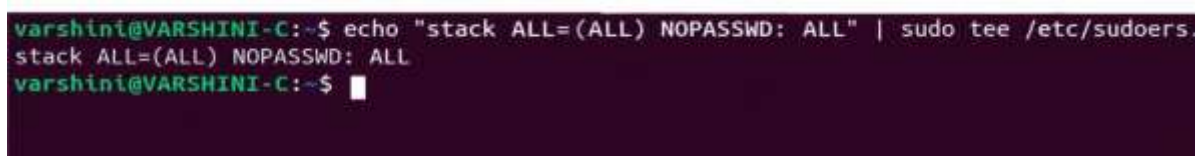
```
varshini@VARSHINI-C:~$ sudo adduser --shell /bin/bash --home /opt/stack stack
[sudo] password for varshini:
Warning: The home dir /opt/stack you specified already exists.
Adding user `stack' ...
Adding new group `stack' (1007) ...
Adding new user `stack' (1004) with group `stack' ...
The home directory `/opt/stack' already exists. Not copying from `/etc/skel'.
adduser: Warning: The home directory `/opt/stack' does not belong to the user you are currently creating.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for stack
Enter the new value, or press ENTER for the default
  Full Name []: KEHSO_STACK
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
varshini@VARSHINI-C:~$
```

4) Set no password for all

A terminal window showing the execution of a command to set no password for all users. The command is 'echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/stack'. The output shows the command being executed successfully.

```
varshini@VARSHINI-C:~$ echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/stack
stack ALL=(ALL) NOPASSWD: ALL
varshini@VARSHINI-C:~$
```

5) Give the privileges for the user stack

A terminal window showing the execution of a command to give privileges for the user stack. The command is 'echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers'. The output shows the command being executed successfully.

```
varshini@VARSHINI-C:~$ echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers
stack ALL=(ALL) NOPASSWD: ALL
varshini@VARSHINI-C:~$
```

6) open the keshoo as sudo user by the command `sudo su - stack`

```
stack@KESHOKUMAR-D:~$ git clone https://git.openstack.org/openstack/devstack
Cloning into 'devstack'...
warning: redirecting to https://opendev.org/openstack/devstack/
remote: Enumerating objects: 52016, done.
remote: Counting objects: 100% (31604/31604), done.
remote: Compressing objects: 100% (10731/10731), done.
Receiving objects: 0% (292/52016), 108.00 KiB | 24.00 KiB/s
```

7) go to home directory and download the openstack from the git by cloning the git .

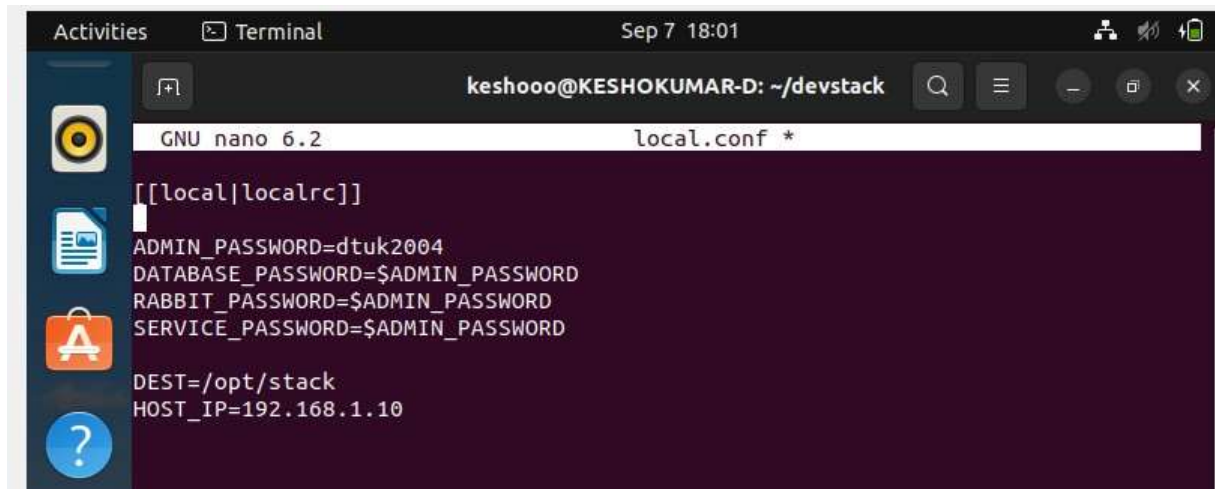
```
stack@KESHOKUMAR-D:~$ git clone https://git.openstack.org/openstack/devstack
Cloning into 'devstack'...
warning: redirecting to https://opendev.org/openstack/devstack/
remote: Enumerating objects: 52016, done.
remote: Counting objects: 100% (31604/31604), done.
remote: Compressing objects: 100% (10731/10731), done.
Receiving objects: 0% (292/52016), 108.00 KiB | 24.00 KiB/s
```

8) Move to the devstack directory

```
varshini@VARSHINI-C:~$ cd devstack
varshini@VARSHINI-C:~/devstack$
```

9) Change the configuration file in the local.conf

Update the below line and give password for each



```

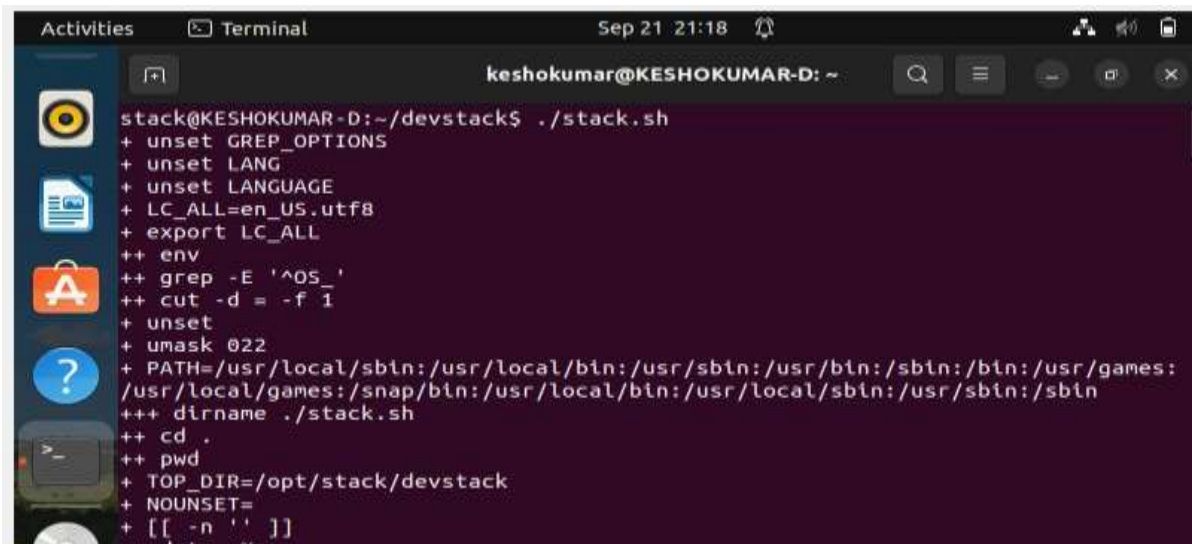
keshoooo@KESHOKUMAR-D: ~/devstack
GNU nano 6.2 local.conf *

[[local|localrc]]

ADMIN_PASSWORD=dtuk2004
DATABASE_PASSWORD=$ADMIN_PASSWORD
RABBIT_PASSWORD=$ADMIN_PASSWORD
SERVICE_PASSWORD=$ADMIN_PASSWORD

DEST=/opt/stack
HOST_IP=192.168.1.10
  
```

10) Run the openstack by the ./stack.sh

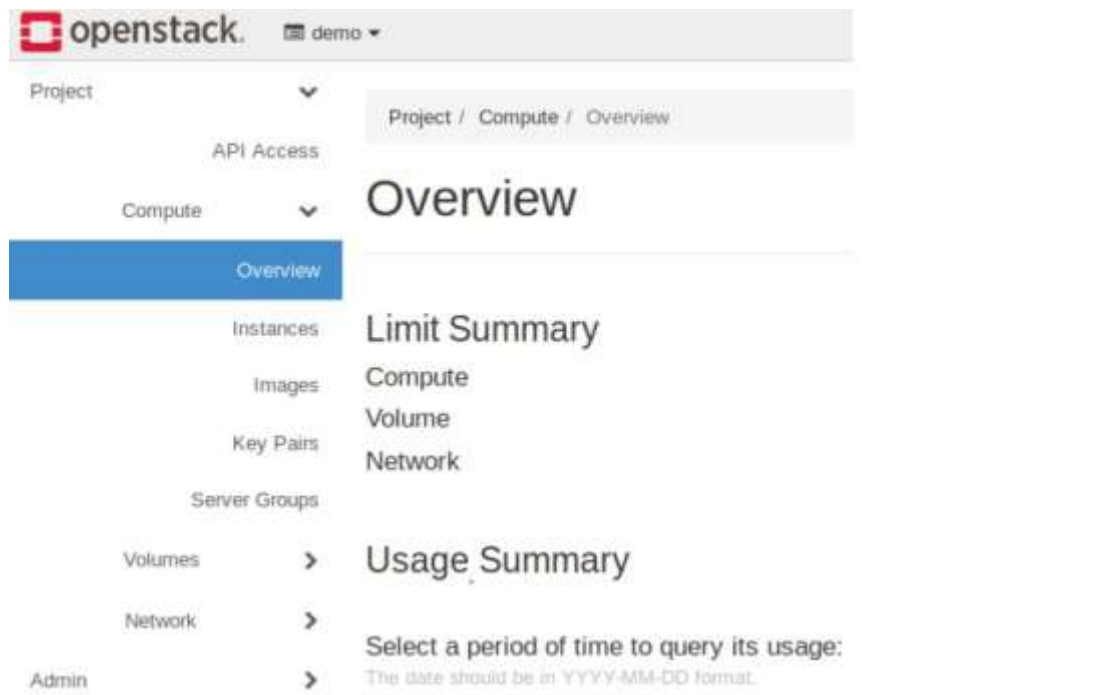


```

keshokumar@KESHOKUMAR-D: ~
stack@KESHOKUMAR-D:~/devstack$ ./stack.sh
+ unset GREP_OPTIONS
+ unset LANG
+ unset LANGUAGE
+ LC_ALL=en_US.utf8
+ export LC_ALL
++ env
++ grep -E '^OS_'
++ cut -d = -f 1
+ unset
+ umask 022
+ PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/usr/local/bin:/usr/local/sbin:/usr/sbin:/sbin
+++ dirname ./stack.sh
++ cd .
++ pwd
+ TOP_DIR=/opt/stack/devstack
+ NOUNSET=
+ [[ -n '' ]]
++ date +%s
  
```

11) Visit the IP address that you have configured in the local.conf and login with the password configured in the local.conf





12) Create a Virtual Machine using openstack glance service using the following command

a) Source OpenStack Credentials.

```
source devstack/openrc admin admin
```

(Replace admin with your actual project and username if different). Verify Image and Flavor.

b) Checkout the image list

```
openstack image list
```

```
openstack flavor list
```

Note the name or ID of the image and flavor you intend to use.

c) Create a Key Pair (if not already done):

```
openstack keypair create mykey > mykey.pem
```

```
chmod 600 mykey.pem
```

d) Create a Network

DevStack typically creates a default network. You can list existing networks:

`openstack network list`. If you need to create a new one, this involves creating a network, subnet, and router, and connecting them. Launch the Virtual Machine.

```
openstack server create --image <image_name_or_id> --flavor <flavor_name_or_id>
--nic net-id=<network_id> --key-name mykey my-first-vm
```

Replace <image_name_or_id>, <flavor_name_or_id>, and <network_id> with your chosen values. my-first-vm is the desired name for your VM. Verify VM Status and IP Address.

```
openstack server list
```

Look for your VM and its assigned IP address. Access the VM.

```
ssh -i mykey.pem cirros@<vm_public_ip>
```

Result:

Thus the installation of openstack and creation of virtual machine in linux system has been executed successfully and output has been verified.